

Teacher's Key • Starters edition

Nota para el docente

Este libro es el *reader* impreso de la Unidad 5 (Grade 3, A1 Movers) y sigue la guía del profesor semana a semana: cada capítulo corresponde a una semana (W1–W11) y el capítulo 5 (volcanes) es lectura de enlace con Science. Las tres ediciones (Starters, Movers, Flyers) tienen los mismos capítulos, imágenes y actividades con distinta dificultad, para agrupar a los alumnos por nivel de lectura sin cambiar de tema. La actividad marcada 🖐 es la entrega oficial de la semana. Cifras verificadas con IGP, INDECI, USGS y NOAA (sept. 2026). El IGP cerró 2025 con 840 sismos reportados.

Chapter 1 • Our Ground Moves

A Drawings and sentences will vary. Check the four words are spelt correctly. • **B** 1c • 2a • 3d • 4b • **C** 1 yes • 2 no • 3 yes • 4 no • **D** Answers depend on the numbers the teacher writes on the diagram. • **E** Answers will vary. • **F** 14

Chapter 2 • Reading the Numbers

A True: 1, 2, 3, 5. (4 — Arequipa; 6 — no, there was an avalanche.) • **B** 1 in 2007 • 2 Arica • 3 five • **C** bigger • **D** 1746 and 1868 • **E** Drawings. • **F** Speaking — no written answer.

Chapter 3 • What Happened in 1970

A 1970 • shakes • fall • run • **B** 2, 4, 3, 1 → order: sunny Sunday, ground shakes, ice falls, people go up the hill • **C** mountain, ice, town, snow • **D** Answers will vary. • **E** Answers will vary.

Chapter 4 • When the Sea Comes

A 2, 5, 3, 1, 4 → 1 ground shakes • 2 walk away • 3 go up • 4 stay • 5 go back when adults say OK • **B** 1 yes • 2 no • 3 yes • **C** Green means safety; the arrow shows the evacuation route up and away from the sea. • **D** ship • **E** Speaking — no written answer.

Chapter 5 • Mountains of Fire

A Drawings will vary. • **B** Sabancaya • **C** 1 yes • 2 no • **D** Example rules: We must wear goggles. We must not put it near our faces. We should do it outside.

Chapter 6 • When People Get Hurt

A e.g. arm / leg • teacher • Thank you • **B** See the body picture in the chapter. • **C** 1a • 2b • 3c • **D** 116 • 106 • **E** Example: “Please come! Diana is hurt. She is in the library. She fell and her arm hurts.”

Chapter 7 • The Backpack

A Answers will vary — check each item is on the INDECI list or is sensible, and the reasons. • **B** ✓ water, torch, whistle, blanket • ✗ TV, football • **C** 7 • **D** Answers will vary — e.g. You must not pack a football. You must put batteries in the torch. • **E** Home task.

Chapter 8 • The Safe Zone and the Drill

A Maps will vary. Safe zone next to columns/under beams, away from windows. • **B** Drop, cover, hold on → walk out → the teacher counts • **C** Game. • **D** Creative. • **E** during • before • after

Chapter 9 • How a Safety Flyer Works

A backpack · hold on · meeting point · **B** Title at the top; three headings (Before, During, After); count the imperatives in the model flyer. · **C** by · **D** A

Chapter 10 • Writing My Flyer

A e.g. torch, whistle, blanket, door · run, knees, head, table · meeting, push, up · **B** must · must not

Chapter 11 • Making It Better

A Final product — assess with the rubric. · **B** Answers will vary. · **C** runs → Don't run / Go under the table (not outside) / Go to the meeting point (not the beach)

Chapter 12 • The Safety Fair

A Personal answers — assess honesty and completeness. · **B** Answers will vary. · **C** Self-assessment.

Teacher's Key • Movers edition

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Chapter 1 • Our Ground Moves

A Drawings and sentences will vary. Check the four words are spelt correctly. • **B** 1b • 2a • 3d • 4c • 5e • **C** 1 T • 2 F (very slowly) • 3 T • 4 F (Germany) • 5 F (centimetres) • **D** Answers depend on the numbers the teacher writes on the diagram. • **E** Answers will vary. • **F** 14 • 60

Chapter 2 • Reading the Numbers

A Answers will vary. Check each fact against the table. • **B** 1 2001 • 2 1868 (Arica) • 3 596 • 4 1746 and 1868 • 5 No — there was an avalanche • **C** 1 stronger than • 2 weaker than • 3 More • 4 Fewer • **D** Lowest: 1970 (7.9). Highest: 1746 and 1868 (about 8.8). • **E** 100 • 1,000 • **F** Speaking — no written answer.

Chapter 3 • What Happened in 1970

A Answers will vary. Check the order, the past simple and the facts. • **B** Order: World Cup (1) • earthquake (2) • ice broke off (3) • people ran up the hill (4) • avalanche covered Yungay (5) • INDECI (6) • **C** started • rushed • climbed • helped • was • were • **D** Answers will vary. • **E** Answers will vary.

Chapter 4 • When the Sea Comes

A See the five steps in the chapter. • **B** 1 F • 2 T • 3 F — it is a warning, go up • 4 T • 5 F • **C** Green means safety; the arrow shows the evacuation route up and away from the sea. • **D** 1 13 August 1868 • 2 about 16 metres • 3 about 430 metres • **E** Speaking — no written answer.

Chapter 5 • Mountains of Fire

A Drawings will vary. • **B** 1 higher than • 2 higher than • 3 The highest • **C** 1 T • 2 F (1600) • 3 T • 4 F (sillar, volcanic stone) • **D** Example rules: We must wear goggles. We must not put it near our faces. We should do it outside.

Chapter 6 • When People Get Hurt

A Answers will vary. • **B** See the body picture in the chapter. • **C** 1b • 2a • 3c • 4d • **D** 1 116 • 2 106 • 3 105 • 4 119 • **E** Example: “Please come! Diana is hurt. She is in the library. She fell and her arm hurts.”

Chapter 7 • The Backpack

A Answers will vary — check each item is on the INDECI list or is sensible, and the reasons. • **B** 1 must • 2 must not • 3 must • 4 must not • **C** 6 kg • yes • **D** Answers will vary — e.g. You must not pack a football. You must put batteries in the torch. • **E** Home task.

Chapter 8 • The Safe Zone and the Drill

A Maps will vary. Safe zone next to columns/under beams, away from windows. • **B** The alarm sounds • We drop, cover and hold on • The shaking stops • We walk out in a line • The teacher counts everybody • We wait for instructions • **C** Game. • **D** Creative. • **E** BEFORE: pack the backpack, learn 119, agree a family meeting point • DURING: drop, cover, hold on; stay away from windows • AFTER: go to the meeting point, check if anyone is hurt, stay away from the beach (on the coast: go up)

Chapter 9 • How a Safety Flyer Works

A Answers will vary. • **B** Title at the top; three headings (Before, During, After); count the imperatives in the model flyer. • **C** 1 Pack a whistle • 2 Stay away from windows • 3 Go to the park • **D** e.g. Pack your emergency backpack today. Choose your family meeting point.

Chapter 10 • Writing My Flyer

A Drafts will vary. • **B** 1 R • 2 A • 3 R • 4 A

Chapter 11 • Making It Better

A Final product — assess with the rubric. • **B** Answers will vary. • **C** must not pack → must pack • Run outside → Don't run outside / Drop, cover, hold on • you should → you must (drop) • Use the lift → Use the stairs • Go to the beach → Go up / Go to the meeting point

Chapter 12 • The Safety Fair

A Personal answers — assess honesty and completeness. • **B** Answers will vary. • **C** Self-assessment.

Teacher's Key • Flyers edition

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Chapter 1 • Our Ground Moves

A Drawings and sentences will vary. Check the four words are spelt correctly. • **B** 1b • 2a • 3d • 4c • 5e • **C** 1 T • 2 F — they get stuck and suddenly slip • 3 T • 4 F — they laughed; proof came in the 1960s • 5 F — there is no proof • **D** Answers depend on the numbers the teacher writes on the diagram. • **E** Answers will vary. • **F** about 16 a week ($840 \div 52 \approx 16.2$) • about 2.3 a day ($840 \div 365 \approx 2.3$)

Chapter 2 • Reading the Numbers

A Answers will vary. Check each fact against the table. • **B** 1 261 years (2007 – 1746) • 2 2007 (8.0 vs 7.9) • 3 1970 — the avalanche and adobe houses • 4 1868 (16 m) and 1746 (10 m) • 5 both caused tsunamis / both hit big cities • **C** Possible ideas: in 2001 it was winter and the beaches were almost empty; 2007 struck at 6:40 pm when a church was full; many old adobe buildings in Pisco collapsed. • **D** The scale is logarithmic: small distances on the chart mean very big differences in energy. • **E** 32 • about 1,000 (1,024) • **F** Speaking — no written answer.

Chapter 3 • What Happened in 1970

A Answers will vary. Check the order, the past simple and the facts. • **B** Order: World Cup (1) • earthquake (2) • ice broke off (3) • people ran up the hill (4) • avalanche covered Yungay (5) • INDECI (6) • **C** /t/ rushed, helped, reached • /d/ buried, travelled, destroyed, covered • /ɪd/ started, shouted, wanted • **D** Answers will vary. • **E** Answers will vary.

Chapter 4 • When the Sea Comes

A See the five steps in the chapter. • **B** 1 F • 2 T • 3 F — it is a warning, go up • 4 T • 5 F • **C** Green means safety; the arrow shows the evacuation route up and away from the sea. • **D** 1 The ship was standing upright on the sand far from the sea • 2 about 15 hours • 3 its rusty boilers • **E** Speaking — no written answer.

Chapter 5 • Mountains of Fire

A Drawings will vary. • **B** Sabancaya is the highest; Ubinas is the most active; Huaynaputina had the biggest eruption; El Misti is the most dangerous for a big city because more than a million people live below it. • **C** 1 T • 2 F (1600) • 3 T • 4 F (sillar, volcanic stone) • **D** Example rules: We must wear goggles. We must not put it near our faces. We should do it outside.

Chapter 6 • When People Get Hurt

A Answers will vary. • **B** See the body picture in the chapter. • **C** 1b • 2a • 3c • 4d • **D** 116 firefighters • 105 police • 106 SAMU ambulance • 115 Defensa Civil • 119 voice messages • **E** Example: “Please come! Diana is hurt. She is in the library. She fell and her arm hurts.”

Chapter 7 • The Backpack

A Answers will vary — check each item is on the INDECI list or is sensible, and the reasons. • **B** 1 rule → You must not make it heavier than 8 kg • 2 advice → You could/should add... • 3 rule ✓ • 4 advice ✓ • **C** 8.3 kg — 0.3 kg too heavy! Remove something. • **D** Answers will vary — e.g. You must not pack a football. You must put batteries in the torch. • **E** Home task.

Chapter 8 • The Safe Zone and the Drill

A Maps will vary. Safe zone next to columns/under beams, away from windows. • **B** The alarm sounds • We drop, cover and hold on • The shaking stops • We walk out in a line • The teacher counts everybody • We wait for instructions • **C** Game. • **D** Creative. • **E** BEFORE: pack the backpack, learn 119, agree a family meeting point • DURING: drop, cover, hold on; stay away from windows • AFTER: go to the meeting point, check if anyone is hurt, stay away from the beach (on the coast: go up)

Chapter 9 • How a Safety Flyer Works

A Answers will vary. • **B** Title at the top; three headings (Before, During, After); count the imperatives in the model flyer. • **C** 1 Use the stairs, not the lift • 2 Go up to high ground at once • 3 Call 119 and leave a message • 4 Put heavy things on low shelves • **D** e.g. Pack your emergency backpack today. Choose your family meeting point.

Chapter 10 • Writing My Flyer

A Drafts will vary. • **B** A rule is necessary for safety (without water you are in danger); advice is a good idea that helps but is not essential.

Chapter 11 • Making It Better

A Final product — assess with the rubric. • **B** Answers will vary. • **C** should not → must not (8 kg rule) • Run to the door → Drop, cover and hold on • Stand in a doorway → Get under a strong desk • Take the lift → Use the stairs • wait for the siren → go up at once • go back after the first wave → wait until the authorities say it is safe

Chapter 12 • The Safety Fair

A Personal answers — assess honesty and completeness. • **B** Answers will vary. • **C** Self-assessment.

Sources and Credits

Sources

- Instituto Geofísico del Perú (IGP) — earthquake reports and catalogue, igp.gob.pe
- INDECI — Combo de la Supervivencia; tsunami recommendations, gob.pe/indeci
- Dirección de Hidrografía y Navegación (DHN) — tsunami warning service
- USGS Earthquake Hazards Program — event pages and “Cool earthquake facts”
- NOAA NCEI — Global Historical Tsunami and Significant Earthquake databases
- USGS Circular 639 (1970): The Peru earthquake of May 31, 1970
- US Navy reports from USS Wateree and USS Powhatan, Arica 1868 (Naval History and Heritage Command)
- Smithsonian Global Volcanism Program — Huaynaputina, El Misti, Ubinas, Sabancaya
- Britannica — Charles F. Richter; Alfred Wegener; Ancash earthquake of 1970
- Minedu / INDECI — Simulacro Nacional Multipeligro 2026
- Nordic International School — Teacher guide, Grade 3, Unit 5: When the Earth Moves

Photo credits

- yungay-1970-avalanche-aerial.jpg — George Plafker, U.S. Geological Survey (NOAA NCEI Natural Hazards Image Database, image 785), Public domain (USGS / NOAA). <https://www.ngdc.noaa.gov/hazardimages/#/all/7/image/785>
- yungay-christ-statue-1970.jpg — U.S. Geological Survey, Public domain (USGS). https://commons.wikimedia.org/wiki/File:Statue_of_Christ_at_Cemetery_Hill_overlooking_old_Yungay.jpg
- yungay-1980.jpg — DB (family archive), Public domain (released by the author). https://commons.wikimedia.org/wiki/File:Yungay,_1980.jpg
- yungay-palms.jpg — Yhhue91, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:Pueblo_de_Yungay_en_%C3%81ncash_Per%C3%BA.jpg
- yungay-christ-statue-today.jpg — Jgaray2000, CC BY-SA 4.0. [https://commons.wikimedia.org/wiki/File:Cristo_de_Yungay_en_el_cementerio_de_Yungay_\(Ancash,_Per%C3%BA\).jpg](https://commons.wikimedia.org/wiki/File:Cristo_de_Yungay_en_el_cementerio_de_Yungay_(Ancash,_Per%C3%BA).jpg)
- huascarán-peaks.jpg — Maurice Chédel, Public domain (released by the author). https://commons.wikimedia.org/wiki/File:Nevado_Huascarán.jpg
- callao-1746-plan.jpg — Unknown 18th-century engraver; reproduced by The West Coast Leader, Ltd. (1925), Public domain. [https://commons.wikimedia.org/wiki/File:\(CCC_No._10._1925\)-_Plano_y_vista_del_Callao,_tal_como_en_el_a%C3%B1o_1746_cuando_fu%C3%A9_destruido_por_el_terremoto.png](https://commons.wikimedia.org/wiki/File:(CCC_No._10._1925)-_Plano_y_vista_del_Callao,_tal_como_en_el_a%C3%B1o_1746_cuando_fu%C3%A9_destruido_por_el_terremoto.png)
- uss-wateree-arica-1868.jpg — Unknown photographer, 1868; Henry Clay Cochrane Collection, USMC Archives (Quantico, USA), CC BY 2.0. [https://commons.wikimedia.org/wiki/File:The_U.S.S._Wateree_\(14037445157\).jpg](https://commons.wikimedia.org/wiki/File:The_U.S.S._Wateree_(14037445157).jpg)
- wateree-boilers-arica.jpg — Pinot (English Wikipedia user), Public domain (released by the author). https://commons.wikimedia.org/wiki/File:Chile_2006_022.jpg
- morro-arica.jpg — Jowyto, CC BY-SA 3.0. https://commons.wikimedia.org/wiki/File:Morro_de_Arica_desde_la_playa.JPG
- arequipa-1868.jpg — Unknown photographer, c. 1868 (Biblioteca Nacional del Perú), Public domain. https://commons.wikimedia.org/wiki/File:Arequipa_devastada_por_el_terremoto_de_1868.jpg
- camana-coast-from-space.jpg — NASA Earth Sciences and Image Analysis Laboratory, Johnson Space Center, Public domain (NASA). [https://commons.wikimedia.org/wiki/File:Tsunami_Vulnerability_in_Camana,_Peru_\(2282\).jpg](https://commons.wikimedia.org/wiki/File:Tsunami_Vulnerability_in_Camana,_Peru_(2282).jpg)
- pisco-2007-after.jpg — Senior Airman Shaun Emery, U.S. Air Force, Public domain (U.S. military). https://commons.wikimedia.org/wiki/File:Pisco_after_the_2007_quake.jpg
- pisco-2007-aid-lima.jpg — Tech. Sgt. Jeremy Lock, U.S. Air Force (via U.S. Army), Public domain (U.S. military). https://commons.wikimedia.org/wiki/File:Aid_after_earthquake_in_Peru.jpg
- misti-arequipa.jpg — Josep M. Gracia, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:ARE_-_El_Misti_volcano_above_Arequipa,_Peru,_2013.jpg
- ubinas-eruption-from-space.jpg — NASA astronaut aboard the International Space Station (NASA Earth Observatory), Public domain (NASA). https://commons.wikimedia.org/wiki/File:Ubinas_ash_cloud_-_ISS.jpg

- sabancaya-eruption.jpg — Ministerio de Defensa del Perú (Flickr gallery), CC BY 2.0.
https://commons.wikimedia.org/wiki/File:Erupci%C3%B3n_en_el_volc%C3%A1n_Sabancaya,_Per%C3%BA.jpg
- huaynaputina-arequipa-1600.jpg — Felipe Guaman Poma de Ayala, Public domain.
https://commons.wikimedia.org/wiki/File:Arequipa_Ascheregen_Huaynaputina_1600.jpg
- plates-map-south-america.jpg — U.S. Geological Survey (cropped by Beyond My Ken), Public domain (USGS).
https://commons.wikimedia.org/wiki/File:South_American_plates.png
- wegener-portrait.jpg — Unknown photographer, Public domain.
https://commons.wikimedia.org/wiki/File:Alfred_Wegener_ca.1924-30.jpg
- charles-richter.jpg — Gil Cooper, Los Angeles Times (UCLA Library Special Collections), CC BY 4.0.
https://commons.wikimedia.org/wiki/File:Charles_Richter_analyzing_a_seismograph_log.jpg
- seismograph-drum.jpg — Z22, CC BY-SA 3.0.
https://commons.wikimedia.org/wiki/File:Seismogram_at_Weston_Observatory.JPG
- zona-segura-sign.jpg — AgainErick, CC BY-SA 3.0 (also GFDL).
https://commons.wikimedia.org/wiki/File:Sign_Peru_Zona_Segura_en_casos_de_sismo.jpg
- tsunami-evacuation-sign-lima.jpg — Gaboct, CC BY-SA 4.0.
https://commons.wikimedia.org/wiki/File:Letrero_sobre_peligro_de_tsunami_en_la_quebrada_de_Armendariz.jpg
- school-drill-lima.jpg — Joaquin Principe, CC BY-SA 4.0.
https://commons.wikimedia.org/wiki/File:Simulacro_nacional_multipeligro_Per%C3%BA_31_mayo_2022_-2.jpg
- lima-balconies.jpg — Felipe Restrepo Acosta, CC BY-SA 4.0. https://commons.wikimedia.org/wiki/File:2017_Lima_-_Balcones_en_el_273_jir%C3%B3n_Ca%C3%B1ete.jpg

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